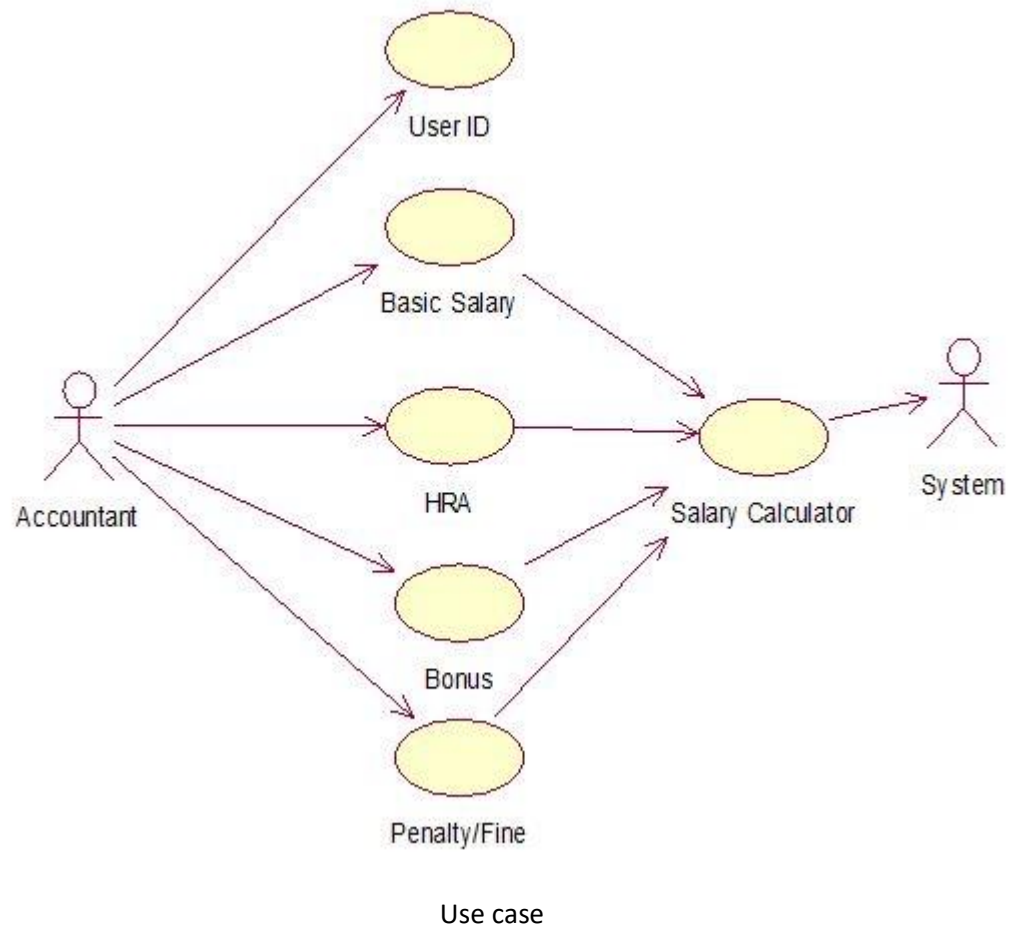
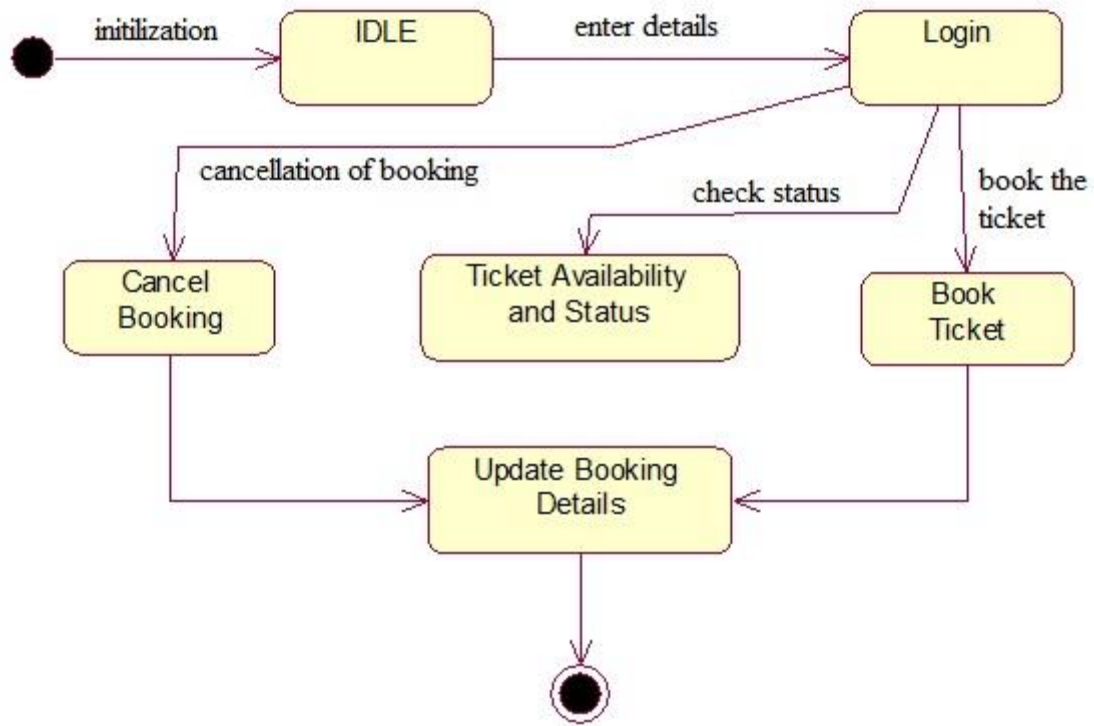


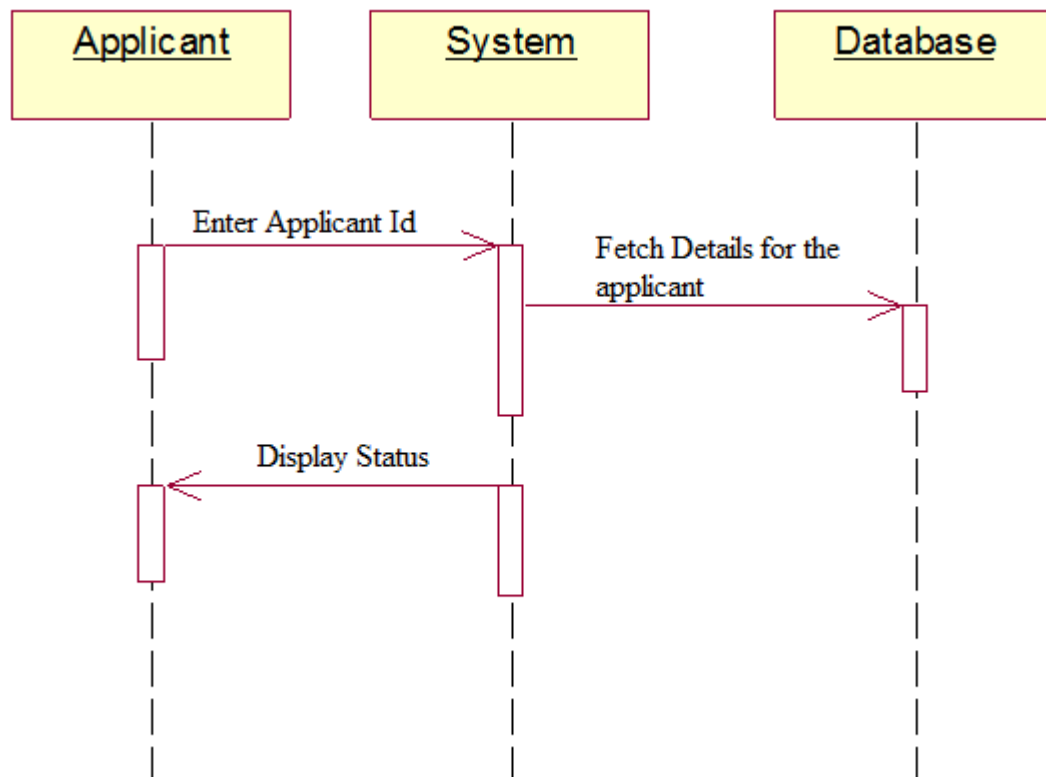
1. How many diagrams are here in Unified Modelling Language?

The nine UML diagrams include use-case, sequence, collaboration, activity, state-chart, deployment, class, object and component.

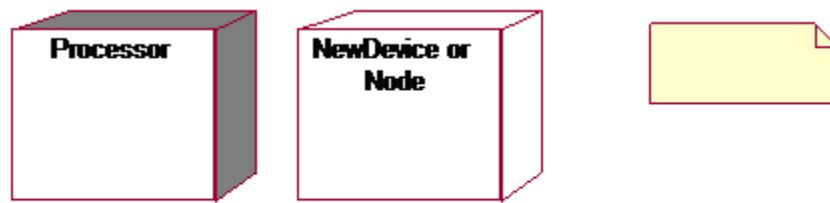




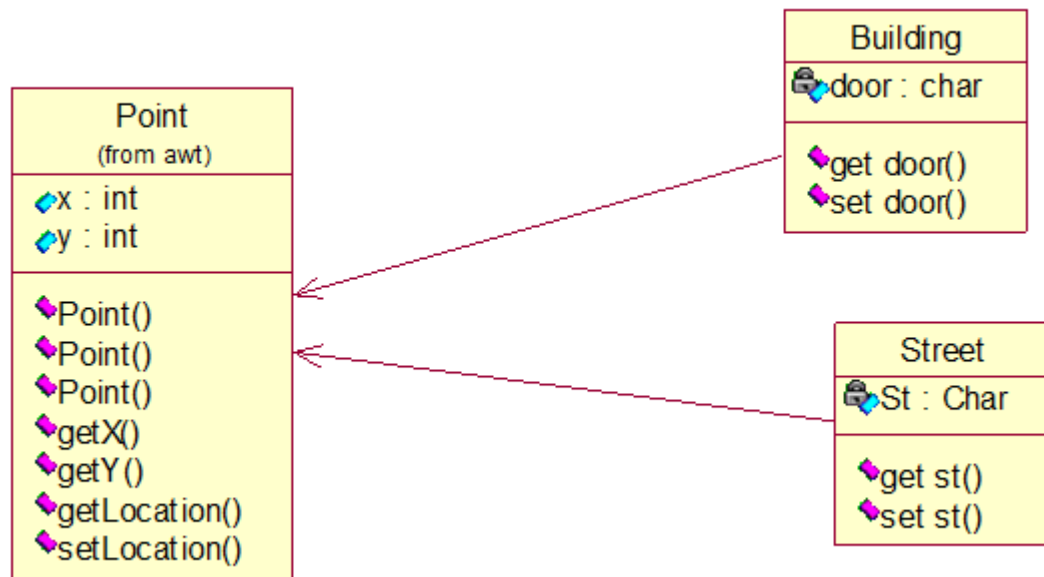
State chart



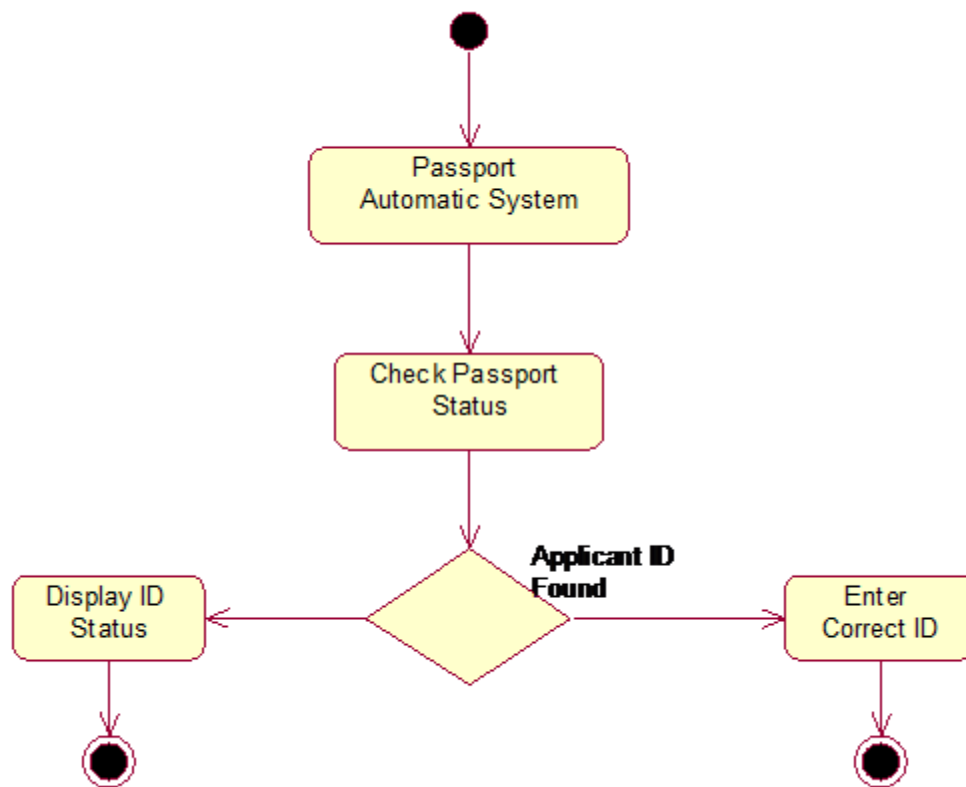
Sequence diagram



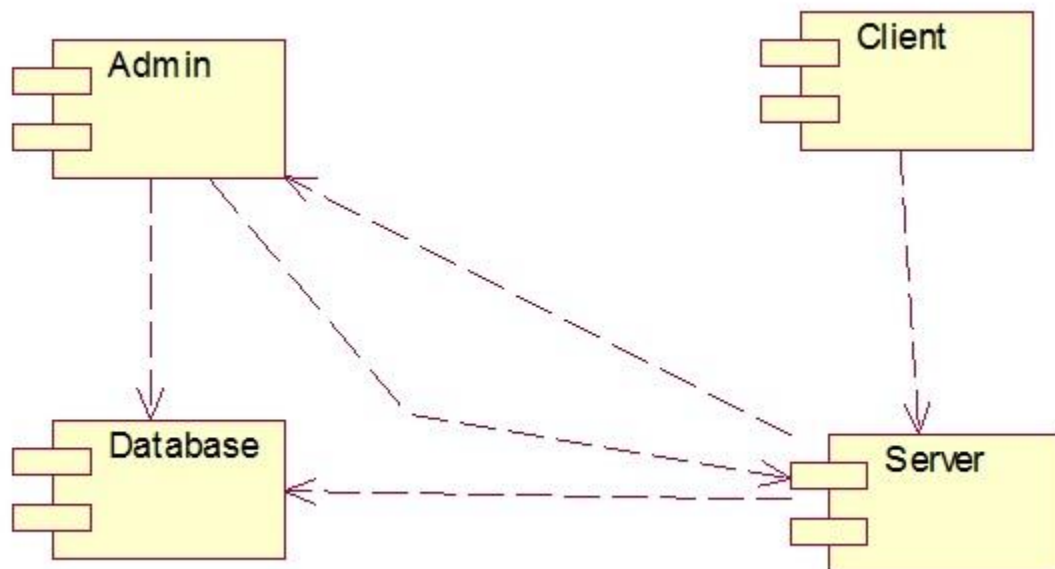
Deployment diagram



Class diagram



Activity diagram



Component

Types of UML Diagrams

The current UML standards call for 13 different types of diagrams: class, activity, object, use case, sequence, package, state, component, communication, composite structure, interaction overview, timing, and deployment.

These diagrams are organized into two distinct groups: structural diagrams and behavioral or interaction diagrams.

Structural UML diagrams

- Class diagram
- Package diagram
- Object diagram
- Component diagram
- Composite structure diagram
- Deployment diagram

Behavioral UML diagrams

- Activity diagram
- Sequence diagram
- Use case diagram
- State diagram
- Communication diagram
- Interaction overview diagram
- Timing diagram

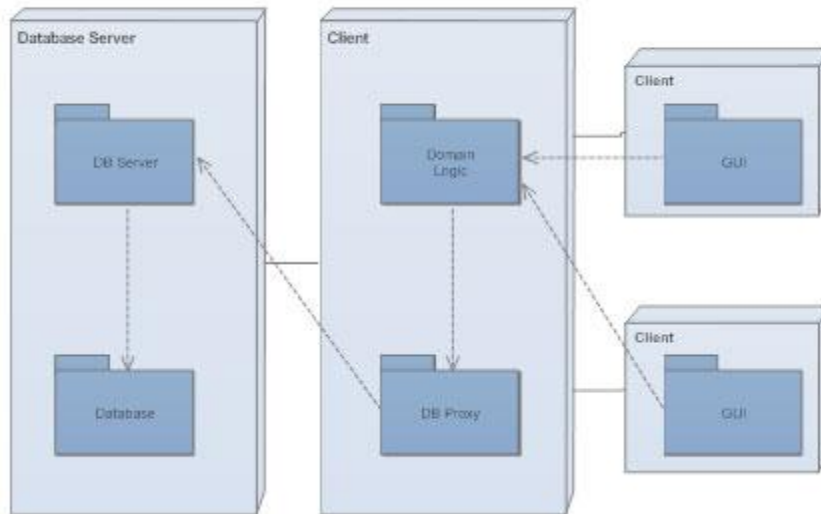
Class Diagram

[Class diagrams](#) are the backbone of almost every object-oriented method, including UML. They describe the static structure of a system. [Learn more](#)

Package Diagram

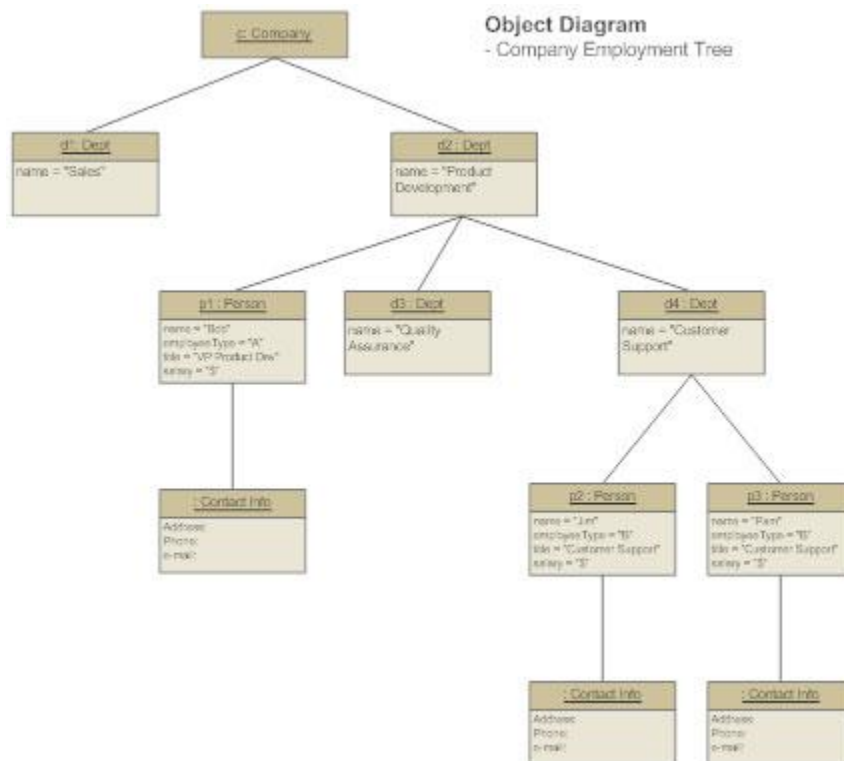
Package diagrams are a subset of class diagrams, but developers sometimes treat them as a separate technique. Package diagrams organize elements of a system into related groups to minimize dependencies between packages.

UML Package Diagram - Encapsulation



Object Diagram

Object diagrams describe the static structure of a system at a particular time. They can be used to test class diagrams for accuracy.

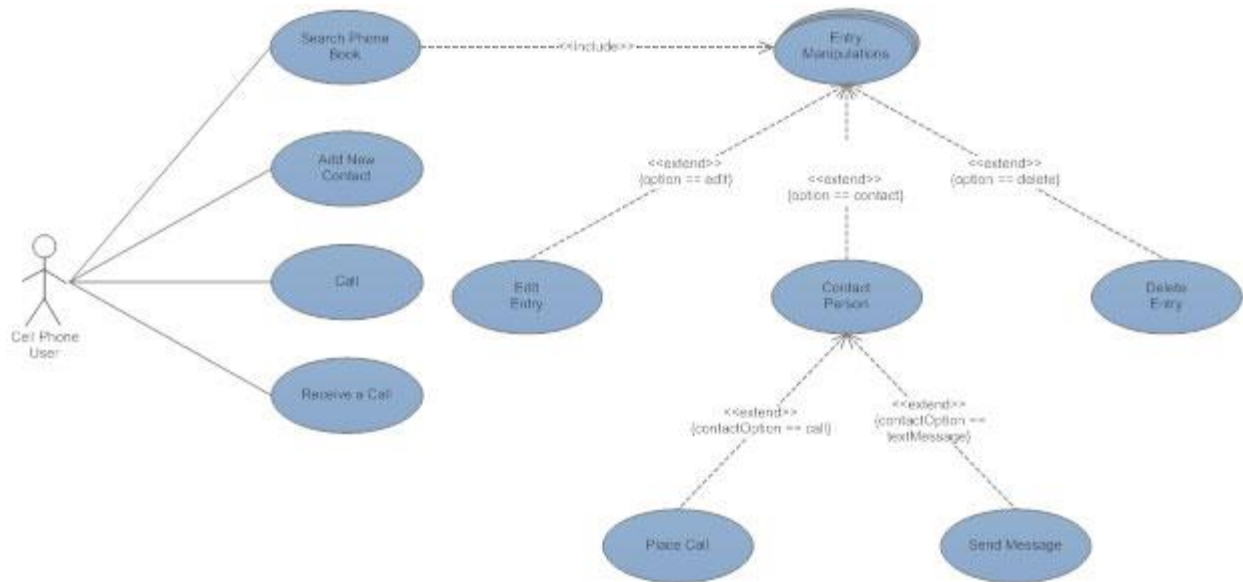


Composite Structure Diagram

Composite structure diagrams show the internal part of a class.

Use Case Diagram

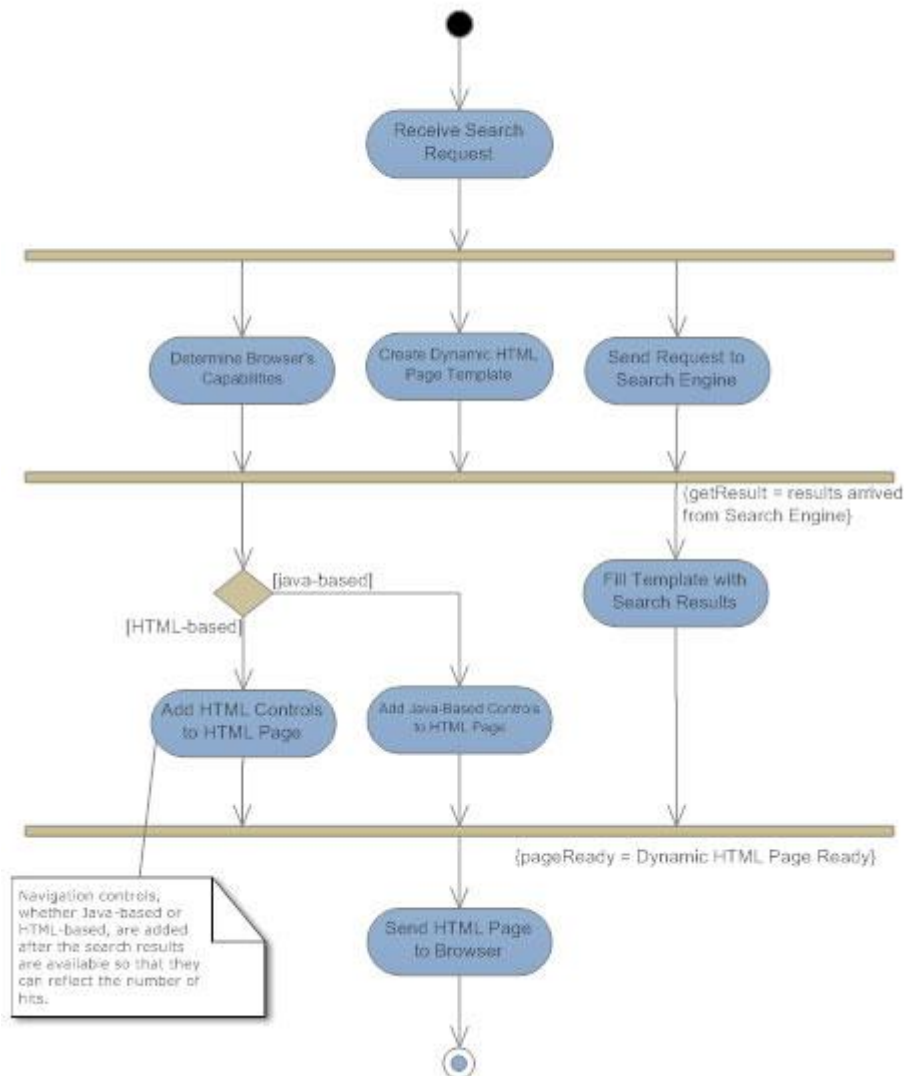
[Use case diagrams](#) model the functionality of a system using actors and use cases. [Learn more](#)



Activity Diagram

[Activity diagrams](#) illustrate the dynamic nature of a system by modeling the flow of control from activity to activity. An activity represents an operation on some class in the system that results in a change in the state of the system. Typically, activity diagrams are used to model workflow or business processes and internal operation. [Learn more](#)

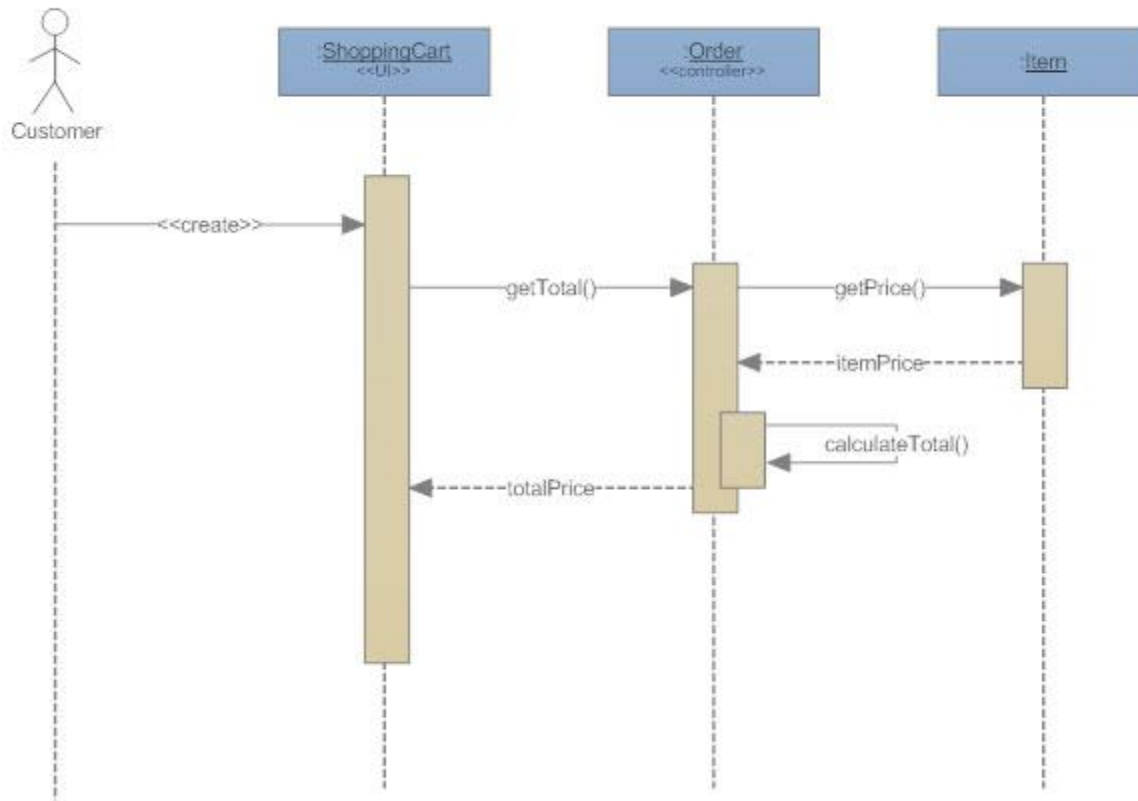
UML Activity Diagram: Web Site



Sequence Diagram

[Sequence diagrams](#) describe interactions among classes in terms of an exchange of messages over time. [Learn more](#)

Sequence Diagram: Shopping Cart



Interaction Overview Diagram

Interaction overview diagrams are a combination of activity and sequence diagrams. They model a sequence of actions and let you deconstruct more complex interactions into manageable occurrences. You should use the same notation on interaction overview diagrams that you would see on an activity diagram.

Timing Diagram

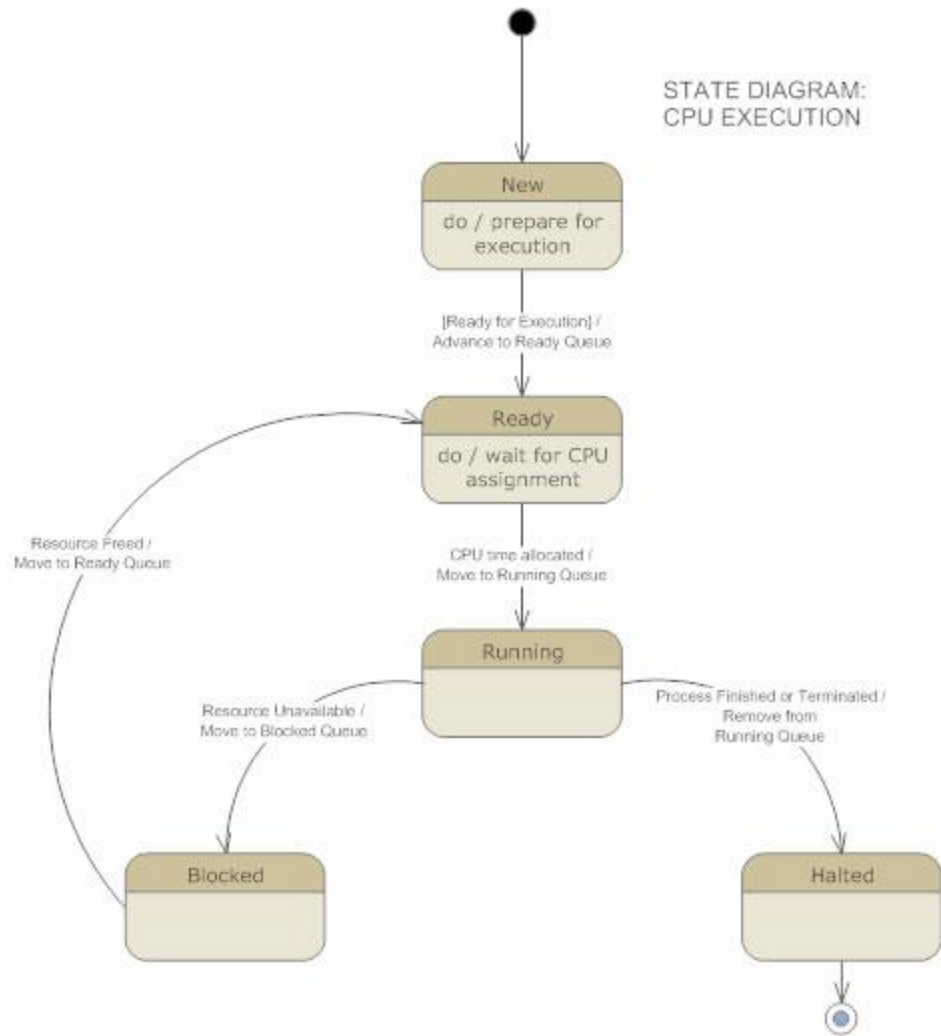
A timing diagram is a type of behavioral or interaction UML diagram that focuses on processes that take place during a specific period of time. They're a special instance of a sequence diagram, except time is shown to increase from left to right instead of top down.

Communication Diagram

Communication diagrams model the interactions between objects in sequence. They describe both the static structure and the dynamic behavior of a system. In many ways, a communication diagram is a simplified version of a collaboration diagram introduced in UML 2.0.

State Diagram

[Statechart diagrams](#), now known as state machine diagrams and state diagrams describe the dynamic behavior of a system in response to external stimuli. State diagrams are especially useful in modeling reactive objects whose states are triggered by specific events. [Learn more](#)

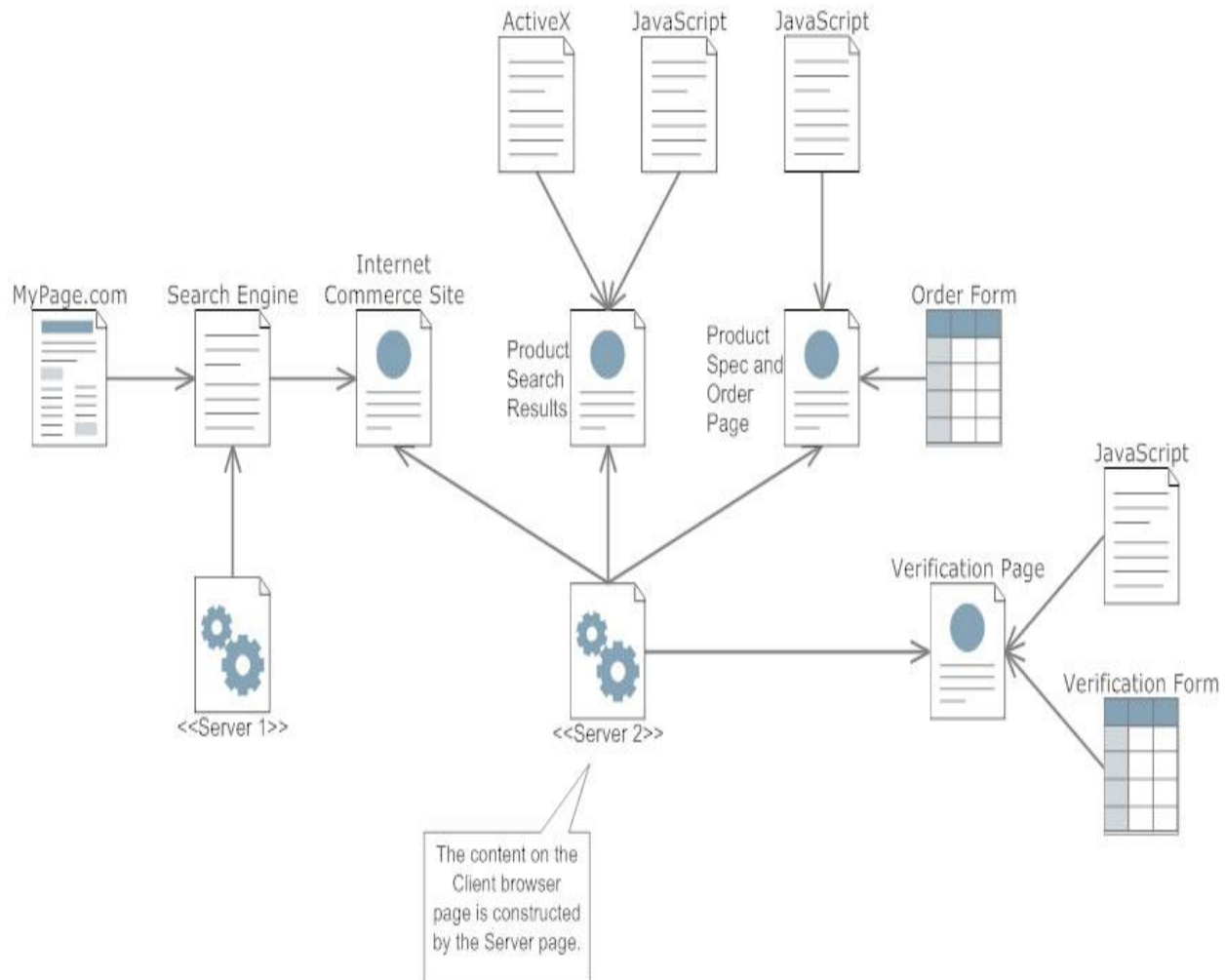


Component Diagram

[Component diagrams](#) describe the organization of physical software components, including source code, run-time (binary) code, and executables. [Learn more.](#)

Web Applications

See Also: UML Class Diagram - Web Transactions



Deployment Diagram

Deployment diagrams depict the physical resources in a system, including nodes, components, and connections.

UML Diagram Symbols

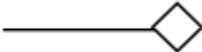
There are many different types of UML diagrams and each has a slightly different symbol set.

Class diagrams are perhaps one of the most common UML diagrams used and class diagram symbols center around defining attributes of a class. For example, there are symbols for active classes and interfaces. A class symbol can also be divided to show a class's operations, attributes, and responsibilities.

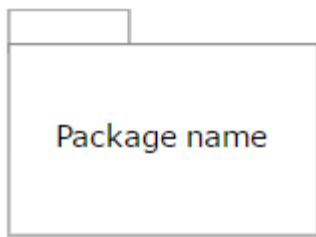
Visibility of any class members are marked by notations of

+	<i>For Public</i>
-	<i>For Private</i>
#	<i>For Protected</i>
/	<i>For Derived</i>
~	<i>For Package</i>

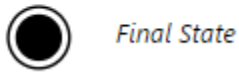
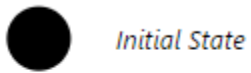
Lines are also important symbols to denote relationships between components. Generalization and Inheritance are denoted with empty arrowheads. Composition is shown with a filled in diamond. Aggregation is shown with an empty diamond. Dependencies are marked with a dashed line with an arrow. Using << >> allows you to indicate properties of that dependency. Multiplicity is usually shown with a number at one end of the arrow and a * at the other.

	<i>Generalization</i>
	<i>Inheritance</i>
	<i>Composition</i>
	<i>Aggregation</i>
	<i>Dependencies</i>
	<i>Properties</i>
	<i>Multiplicity</i>

Package diagrams have symbols defining a package that look like a folder.



Activity diagrams have symbols for activities, states, including separate symbols for an initial state and a final state. The control flow is usually shown with an arrow and the object flow is shown with a dashed arrow.



Use case diagrams have symbols for actors and use cases